

Question_6:

Code:

```
1  /*
2  Question_6:
3  Perform Insertion Sort, print the array after the
4  3rd iteration and then print the final sorted array.
5  */
6
7  #include <iostream>
8  using namespace std;
9
10 void printArray(int arr[], int n) {
11     for (int i = 0; i < n; i++)
12         cout << arr[i] << " ";
13     cout << endl;
14 }
15 void insertionSort(int arr[], int n) {
16     for (int i = 1; i < n; i++) {
17         int x = arr[i], j = i - 1;
18
19         while (j >= 0 && arr[j] > x) {
20             arr[j + 1] = arr[j];
21             j--;
22         }
23         arr[j + 1] = x;
24         if (i == 2)
25             printArray(arr, n);
26     }
27     printArray(arr, n);
28 }
29 int main() {
30     int n;
31     cin >> n;
32
33     int arr[n];
34     for (int i = 0; i < n; i++)
35         cin >> arr[i];
36     insertionSort(arr, n);
37     return 0;
38 }
```

Output:

Test Case_1:

```
5
64 25 22 90 35
22 25 64 90 35
22 25 35 64 90
PS C:\Users\Prabin Yadav\I
```

Test Case_2:

```
5
16 21 22 10 35
16 21 22 10 35
10 16 21 22 35
PS C:\Users\Prabin Yadav\I
```